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Fire, Flood and Cyclone: Living with Hazards in Western Australia (Geography — Landforms and landscapes: hazards)

TEACHER COPY — NOT FOR STUDENTS

Worksheet

Part A

1. e.g. *Police and emergency services evacuated hundreds of homes overnight.* (Any plausible agent; the active version must name it.)
2. e.g. *Climate change is **likely** to make **many** WA summers more dangerous.* / *...**may** make summers in **parts of** WA more dangerous.* Two hedges needed (modal/adverb + limiting quantifier or scope).
3. a) emotive language / appeal to fear (pathos) b) statistics / appeal to the majority (bandwagon) c) rhetorical question (also inclusive *we*)
4. Any two: Who posted it, and how would they know? Is there an official source (police, DFES) confirming it? Is it a primary source (eyewitness) or repeating a rumour? Does the poster have a motive? Has any reliable outlet reported it?

Part C

1. **Destructive winds** — strip roofs, turn loose objects into missiles. **Storm surge** — sea rises above high tide, floods low-lying coasts. **Heavy rain** — floods, including far inland.
2. ① Fitzroy River flood (Fitzroy Crossing), flood after ex-Tropical Cyclone Ellie, Dec 2022 – Jan 2023 · ② Severe Tropical Cyclone Seroja (Kalbarri), cyclone · ③ Wooroloo bushfire (Perth Hills), bushfire.
3. S in **April**, W in **February**, F in **December or January** (accept either; the rain began in late December and the flood peaked in early January). The calendar shows the northern cyclone/wet season and the south-west bushfire season overlap almost exactly (roughly Nov–Apr), so DFES and volunteers may be dealing with cyclones or floods in the north and fires in the south **at the same time**, stretching people and equipment.
4. A hazard is a natural event that *could* cause harm; it becomes a disaster when it hits people and infrastructure that are exposed and unprepared. Seroja's winds were the hazard; it became a disaster in Kalbarri and Northampton because homes there had not been built for cyclone winds, so many were damaged.
5. Weather: it kept Category 3 strength unusually far south. People's decisions: building rules for that area did not require cyclone-resistant construction, so homes weren't designed for those winds (planning relied on past patterns).
6. Most strong answers agree. Evidence: Seroja (¶5) — a forecast cyclone still did heavy damage because buildings were unprepared; ¶6 — rating system and warnings only help if people act (*plan and prepare*, *leave*); ¶6 volunteers; ¶3 community organisations and local knowledge in recovery;

¶7 safety depends on planning and neighbours. A sophisticated answer notes prediction and preparation depend on each other (warnings give time to act on plans).

Connections

7. Port Hensley: 178 km/h → **Category 3**. Spinifex Point: 141 km/h → **Category 2** gusts.
8. Wind falls steadily inland (178/141 on the coast → 87 → 52 km/h). Rain does **not** fall steadily: Cooper Creek (164 mm) and Red Hill (118 mm) had more rain than coastal Spinifex Point (96 mm). Explanation (¶2): over land the cyclone loses its supply of warm, moist air and weakens, so winds drop, but its remains still carry moisture that falls as rain hundreds of kilometres inland — hence inland flood risk.
9. a) (212 + 96 + 164 + 118) ÷ 4 = 590 ÷ 4 = **147.5 mm**. b) Only **one** of the four stations (Port Hensley) recorded more than 200 mm; "across the region" over-generalises. Model: *According to the Bureau of Meteorology, 24-hour rainfall ranged from 96 mm at Spinifex Point to 212 mm at Port Hensley, with inland stations recording more than 100 mm.* (Accept a version giving the mean, as long as it is attributed and accurate.)
10. 178 ÷ 1.852 = 96.1… → **96 knots**.
11. a) **30%** (complementary event: 100% − 70%). b) The headline turns a probability into a certainty; readers may panic or, if gales don't arrive, stop trusting forecasts. Model: *Gales likely at Red Hill within 24 hours, Bureau says* / *Red Hill faces 70% chance of gales*.

Part D

12. Tier 1 = **lead** (¶1). Tier 2 = **key facts / official details and evidence** (¶2–¶3). Tier 3 = **further detail, human-interest quotes, what happens next** (¶4–¶6). Tier 4 = **background** (¶7). Accept ¶6 in tier 2 or 3 with a reason. ¶7 is cut first because it is background — the story still makes complete sense without it.
13. Who: Tropical Cyclone Harlow (and residents) · What: crossed the coast as a Category 3, damaged roofs and power lines · Where: Pilbara coast near Port Hensley · When: about 2 am today. **Why** is not answered (and **how** only partly): the cause of a cyclone is not news to readers, and in a breaking report the facts of what happened matter more than explanation.
14.

Report A	Report B	Effect
damaging roofs	*Homes were ripped apart* / *tore through* / *trail of devastation*	violent verbs exaggerate damage; create fear and shock
a gust of 178 km/h	*winds of almost 200 km/h*	rounds **up** to a dramatic figure, drops the source and the word *gust* (a gust is a short burst, not a steady wind)
about 40 homes had reported damage	*dozens of homes have been destroyed*	changes *damaged* to *destroyed*; vague *dozens* and unsourced *It is understood* make it sound worse than the evidence

15. Any two: no serious injuries; Category 3 and the official gust figure; the DFES safety advice; about 120 people sheltered in the evacuation centre; residents acted on the Red Alert; the cyclone is

expected to weaken. Omitting them removes the good news and the evidence that warnings and preparation worked, so the event seems more catastrophic and authorities seem to have failed.

16. A: Bureau of Meteorology, DFES (and a spokesperson), resident Tom Walsh, Shire President Dana Ruiz, unnamed residents (criticism). B: one resident (Kylie Brant), unnamed “furious locals”, “experts”. Vague phrases hide who is making a claim, so readers cannot check the source, judge its expertise, or know if it exists at all; they let the writer slip in unsupported claims (*dozens destroyed*, *wall of water*) while sounding authoritative.
17. Nobody is named as asking; no one is named as responsible (*whether enough was done* — by whom?). The passive removes the agent, so the writer can imply authorities failed without naming them, providing evidence or giving them a right of reply. An active version would have to say *who* asks and *whom* they blame — which would require a source.
18. Selection: choosing the most emotional quote from one person and leaving out others (e.g. those who acted on warnings) creates an unrepresentative picture. Framing: *sobbed* and *clutching her youngest child* add emotion that the reporter, not the source, chose. Placement high in the story gives it extra weight. (Note Report A also includes a criticism in ¶5 — but balanced and lower down.)

Part E

19. The belief that the positions of stars and planets (astrology) caused bad luck or misfortune — a disaster was something “ill-starred”. The text instead explains disasters as caused by natural hazards meeting **exposed, unprepared** people: a human and geographical explanation, not fate.
20. *Damage* — it is measurable and neutral; *catastrophe* is evaluative and emotive. *Catastrophe* may be accurate when the scale is extreme and documented (e.g. when officials use it, in an attributed quote), or in analysis/feature writing rather than the reporter’s own voice.

Part F — marking notes

Model report (≈280 words)

Three homes lost as Jarrah Ridge fire downgraded

A bushfire destroyed three homes and seven sheds near the hills town of Jarrah Ridge, about 70 km south-east of Perth, on Saturday, before the warning was downgraded to Watch and Act at 9 pm.

According to the Department of Fire and Emergency Services (DFES), the fire started in bushland north of the town about 11:30 am and had burnt about 1200 hectares by 8 am Sunday. An Emergency Warning was issued at 1:15 pm on a day rated Extreme, with temperatures reaching 41 °C.

DFES Incident Controller Priya Nair said a south-westerly wind change about 4 pm "turned the eastern flank into a new front", but crews held the line on Ridge Road, "which protected more than 200 homes".

Two volunteer firefighters were treated for heat exhaustion and released from hospital on Saturday night. Brigade captain Sam O'Brien said crews had worked "14 hours straight".

About 300 people sheltered at the evacuation centre at the Jarrah Ridge Recreation Centre.

Resident Mel Hart, whose home was destroyed, said her family left at 1 pm after following their bushfire plan. "We lost the house, but we're all safe," she said.

The cause of the fire is under investigation. Police have not commented on claims circulating on social media.

The last major fire near Jarrah Ridge was in 2011.

Criterion	Developing (1)	Sound (2)	Strong (3)
Headline	missing or sensational	accurate but long or vague	short, accurate, present tense, most newsworthy fact
Lead	fewer than three of the 5Ws + H, or buried detail	four of the 5Ws + H	four or more, concise, in 1–2 sentences, no background
Structure (inverted pyramid)	chronological or random	mostly ordered by importance	clearly most → least important; background last; could be cut from the bottom
Attribution and sources	claims unattributed	most figures attributed	every figure/claim attributed; ≥2 different kinds of source (official + community/volunteer)
Objectivity	emotive adjectives in reporter's voice; rumour reported as fact	mostly neutral; minor lapses	neutral throughout; emotion only inside attributed quotes; rumour omitted or clearly marked unconfirmed
Accuracy and hedging	figures changed or invented	figures correct; few hedges	all figures exactly as in fact file; hedges where needed (*about*, *under investigation*)
Language control	errors impede meaning	mostly accurate; some long sentences unclear	precise, varied sentences; correct quote punctuation; 250–300 words

Total: /21 (7 criteria × 3). Look especially for: whether the student **selected** (a strong report does not use every fact), correct punctuation of quotes, and whether *destroyed* vs *damaged* is kept accurate. Common errors: starting with the time of ignition (chronological), writing “devastating fire”, reporting the social-media claim as fact, quoting only the resident.

Part G

21. Choices that shape the view: the order puts *No serious injuries* in the lead and ends the official section with the Shire President’s positive quote, framing the response as a success; the vivid freight-train simile from Tom Walsh adds drama; official sources outnumber residents; the criticism about storm-surge information is unattributed (*some residents*) and placed after the Shire President. Strong answers conclude that it is still **fair** because claims are attributed, figures are exact, criticism is included and the reader can judge — objectivity is a matter of degree.
22. In Report A the agent is either unknown or obvious (no injuries reported *to authorities*), the claim is factual and time-limited, and nothing is being blamed. In Report B ¶6 the passive hides the people making an accusation and the people being accused, so it insinuates blame without evidence.

Fast-finisher bank — answers & marking notes

F1. Model (≈70 words): *Tropical Cyclone Harlow crossed the coast near Port Hensley overnight, damaging homes and bringing down power lines. The Bureau of Meteorology recorded a gust of 178 km/h. Resident Kylie Brant, a mother of two, said it was "the worst night of my life" and that she had received little warning. No serious injuries had been reported by Tuesday morning.* Check: neutral verbs (*said*), exact figure, balance (the key safety fact restored).

F2. e.g. Neutral: *Wooroloo bushfire destroys more than 80 homes in Perth Hills*. Sensational: *INFERNO! Hills homes reduced to ash*. Question: *Could your street be next?* Clicks: sensational/question (curiosity, fear); trust: neutral (specific, verifiable, no exaggeration).

F3. e.g. *How has the community worked together since the flood? / What has been the hardest part of the recovery? / What do you want people outside the Kimberley to understand? / How has the bridge closure affected daily life? / What has helped most, and what is still needed?* Open questions (*how, what, why*) invite explanation and personal detail; closed ones get yes/no answers. Reward respectful framing (asking what the community wants said, not sensational questions about loss).

F4. a) Loaded verb *slam* (conflict framing) + emotive *chaos*; capitals for emphasis; makes a dispute seem dramatic and the council guilty. b) Direct address *YOUR* + rhetorical question; appeals to fear and self-interest; clickbait — the answer is inside the article. c) Neutral "explainer" pattern; signals provisional, factual information and openly admits uncertainty — the most objective of the three.

F5. Model: **Cyclone Seroja crosses coast near Kalbarri** — *Severe Tropical Cyclone Seroja crossed the coast near Kalbarri, about 500 km north of Perth, on Sunday night as a Category 3 system, damaging buildings in Kalbarri and Northampton. Power lines were brought down across the Mid West. Cyclones rarely keep this strength so far south, and many homes in the area were not built to cyclone standards.* (11 April 2021 was a Sunday.) Penalise any invented figures (deaths, number of homes, wind speeds).

F6. Strong responses balance public interest (showing the human cost, prompting aid and accountability) with dignity, consent and harm (people in shock may not be able to consent; images circulate permanently). For many Aboriginal communities, showing images, names or voices of people who have died can cause distress, and media outlets commonly publish warnings or seek family/community permission; good journalism consults the community. Reward a clear position with a concession.

F7. Look for: a narrative hook (scene, character) that is plausible and not sensational; accurate information (season 1 Nov – 30 Apr; Broome–Exmouth coast most cyclone-prone; three dangers; Blue/Yellow/Red alerts; volunteers/SES); no invented statistics; shift in register from news (descriptive, personal angle permitted).

F8. Strong responses: concede that every report involves selection (*Although even Report A chooses which quotes to include and in what order…*), argue that fairness (accuracy, attribution, balance of voices, right of reply, separating fact from opinion) is achievable and measurable; Report B fails fairness tests (vague sources, omission, hidden agency), not merely objectivity. A sophisticated answer may rebut the claim that "everything is biased so all reports are equal" — degrees of fairness matter.

Homework

1. A eye · B eyewall · C spiral rainband · D warm moist air spiralling in · E air spreading outwards at the top.

2. (1) Warm water evaporates rapidly, supplying moist air. (2) When that vapour condenses in the clouds it releases heat, which makes the air rise faster and draws in more moist air — a self-strengthening cycle.
3. The calm eye is temporary; after minutes to an hour or more, the eyewall arrives and violent winds return from the opposite direction, catching people outside and possibly damaging structures already weakened from the other side.
4. Fuel = heat and moisture (water vapour) from the warm ocean. Breakdowns: an engine is designed and controlled and can be switched off; a cyclone has no fixed parts; engines usually burn fuel chemically, whereas a cyclone gains energy from condensation. Any reasoned limitation.
5. *evaporation* (from *evaporate*), *condensation* (from *condense*) (also accept *moisture* from *moist* with explanation). e.g. *Condensation releases heat → When water vapour condenses, it releases heat.* The nominalised version is more concise and impersonal, suiting scientific register; the verb version is clearer for younger readers.
6. Emotive/misleading: *deadly* (an eye has light winds; the danger is the returning eyewall), *tricks* (personification implying intent). Model: *The calm eye of a cyclone can lead people to go outside before strong winds return.*
7. a) All contain the idea of a circle/wheel: *bicycle* — two wheels; *recycle* — go round again; *encyclopaedia* — an "all-round" (circle of) education. b) A cyclone is a circular, rotating system; the diagram shows air circulating in, up, out and down around the eye. c) The Kimberley's weather is dominated by one wet season and one dry season, so the rain arrives as a *season* — the defining feature of a monsoonal climate.
8. Any genuine response. Check the news brief: neutral, attributed opinion (*Mum said…*), no emotive language. Warning levels: **Advice** — an incident has started; stay up to date. **Watch and Act** — conditions are changing; take action now to protect yourself and your family. **Emergency Warning** — you are in danger and need to take action immediately (highest level).

Teacher notes

- **Program link:** Week 31 introduces news structure and objectivity; Week 32 (*Same Event, Different Story*) deepens bias, framing and omission. Keep Reports A and B — they can be reused as a warm-up in Week 32.
- **Common misconceptions:** thinking "objective" means "no quotes with emotion" (emotion in an attributed quote is fine; emotion in the reporter's own voice is not); thinking a true quote cannot be biased; treating all passive voice as biased; confusing *gust* with sustained wind; chronological rather than inverted-pyramid structure.
- **Sensitivity:** students may have family affected by the Wooroloo fire, Seroja or the Kimberley floods. Check before the lesson. The text refers to Aboriginal communities respectfully and in general terms; avoid speculating about specific communities. *Martuwarra* is the name used by many Traditional Owners for the Fitzroy River (e.g. by the Martuwarra Fitzroy River Council).
- **If the student finishes early:** have them find a real news report about a recent hazard and annotate it with the six bias tools from Part D.
- **If the student struggles:** do Q14 orally first; highlight every attribution verb in Report A in one colour and every emotive word in Report B in another.

Sources / fact-check notes

Facts in the main text are deliberately rounded and limited to well-documented points. Verify against these if challenged:

- **WA area ≈ one-third of Australia** (about 2.5 million km²); north–south extent well over 2000 km — Geoscience Australia.
- **Cyclone formation** generally requires sea-surface temperatures of about 26.5 °C or more; Australian season officially 1 November – 30 April; **Pilbara coast between Broome and Exmouth** described by the Bureau of Meteorology as the most cyclone-prone part of the Australian coast — BoM tropical cyclone pages.
- **Cyclone category scale** (strongest gust): Cat 1 < 125 km/h, Cat 2 125–164, Cat 3 165–224, Cat 4 225–279, Cat 5 ≥ 280 — BoM. 1 knot = 1.852 km/h (exact).
- **Ex-Tropical Cyclone Ellie / Fitzroy River flood:** Ellie crossed the NT coast in late December 2022 and its remnant low lingered over the Kimberley into early January 2023; record flood at Fitzroy Crossing; Great Northern Highway Fitzroy River bridge damaged beyond repair, replaced (new bridge opened late 2023); temporary crossing built in the meantime — BoM, WA Government/Main Roads WA media statements. The reported peak was about 15.8 m at the Fitzroy Crossing gauge (previous record about 13.95 m, 2002) — **left out of the student text** because exact gauge figures vary between sources.
- **Wooroloo bushfire:** started 1 February 2021; 86 homes destroyed; about 10 900 ha burnt; Shire of Mundaring and City of Swan; Perth was in a five-day snap COVID-19 lockdown that began 31 January 2021 — DFES, Independent review / WA Government statements. Text says "more than 80 homes" and "roughly 11 000 hectares".
- **Severe Tropical Cyclone Seroja:** crossed the coast near Kalbarri on the evening of 11 April 2021 as a Category 3; severe damage in Kalbarri and Northampton; widespread power outages in the Mid West; area is in a non-cyclonic wind region of the building code, which was widely discussed after the event (e.g. Cyclone Testing Station, James Cook University, damage report) — BoM, DFES. Seroja had earlier caused deadly floods in Indonesia and Timor-Leste; this is not in the student text.
- **Australian Fire Danger Rating System:** introduced nationally 1 September 2022; levels Moderate, High, Extreme, Catastrophic; key messages "Plan and prepare", "Be ready to act", "Take action now to protect your life and property", "For your survival, leave bushfire risk areas" — AFAC / DFES.
- **Australian Warning System:** Advice, Watch and Act, Emergency Warning. WA cyclone alerts: Blue, Yellow, Red, All Clear — DFES / Emergency WA.
- **Climate trends:** south-west WA April–October rainfall has declined since the 1970s (roughly 15–20 %); increase in dangerous fire-weather days and longer fire seasons across southern Australia; projections of fewer but proportionally more intense tropical cyclones (medium/lower confidence) — BoM & CSIRO *State of the Climate* reports (2020, 2022, 2024). Percentages deliberately left out of the student text.
- **Volunteers:** DFES supports tens of thousands of emergency services volunteers; the text says "thousands", which is safely conservative.
- **Homework:** Coriolis effect and hemisphere rotation, eye/eyewall/rainband structure, weakening over land — BoM; *cyclone* coined by Henry Piddington in the 1840s (Oxford English Dictionary/etymology dictionaries give 1848); *monsoon* from Portuguese *monção* < Arabic *mawsim*; *tropic* from Greek *tropē*.
- **Everything about Cyclone Harlow, Port Hensley, Spinifex Point, Cooper Creek, Red Hill and Jarrah Ridge is fictional**, including all people quoted.